

Step 9: Data Visualization (Matplotlib & Seaborn)

Visualizing the key findings from the SQL and Python analysis:

1. Monthly revenue trend (seasonality)
2. Profit by category (highlighting the Furniture margin issue)
3. Discount band vs average profit
4. Regional sales distribution
5. Profit margin distribution (outliers)
6. Top 10 products by sales

```
In [21]: import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

df = pd.read_csv('../data/superstore_cleaned.csv')
df['OrderDate'] = pd.to_datetime(df['OrderDate'])
df['ShipDate'] = pd.to_datetime(df['ShipDate'])

sns.set_style('whitegrid')
plt.rcParams['figure.figsize'] = (10, 6)

print("Data loaded:", df.shape)
df.head()
```

Data loaded: (9993, 26)

Out[21]:

	RowID	OrderID	OrderDate	ShipDate	ShipMode	CustomerID	CustomerName	Segment
0	1	CA-2016-152156	2016-11-08	2016-11-11	Second Class	CG-12520	Claire Gute	Consumer
1	2	CA-2016-152156	2016-11-08	2016-11-11	Second Class	CG-12520	Claire Gute	Consumer
2	3	CA-2016-138688	2016-06-12	2016-06-16	Second Class	DV-13045	Darrin Van Huff	Corporate
3	4	US-2015-108966	2015-10-11	2015-10-18	Standard Class	SO-20335	Sean O'Donnell	Consumer
4	5	US-2015-108966	2015-10-11	2015-10-18	Standard Class	SO-20335	Sean O'Donnell	Consumer

5 rows × 26 columns



Chart 1: Monthly Revenue Trend

Shows seasonality — revenue peaks in November and December each year.

```
In [22]: monthly_revenue = df.groupby(['OrderYear', 'OrderMonth'])['Sales'].sum().reset_index()
monthly_revenue['Period'] = monthly_revenue['OrderYear'].astype(str) + '-' + monthly_revenue['OrderMonth'].astype(str)

plt.figure(figsize=(14, 6))
plt.plot(monthly_revenue['Period'], monthly_revenue['Sales'], marker='o')
plt.title('Monthly Revenue Trend (2014-2017)')
plt.xlabel('Month')
plt.ylabel('Total Revenue ($)')
plt.xticks(rotation=90)
plt.tight_layout()
plt.savefig('../dashboard/chart1_monthly_revenue.png')
plt.show()
```

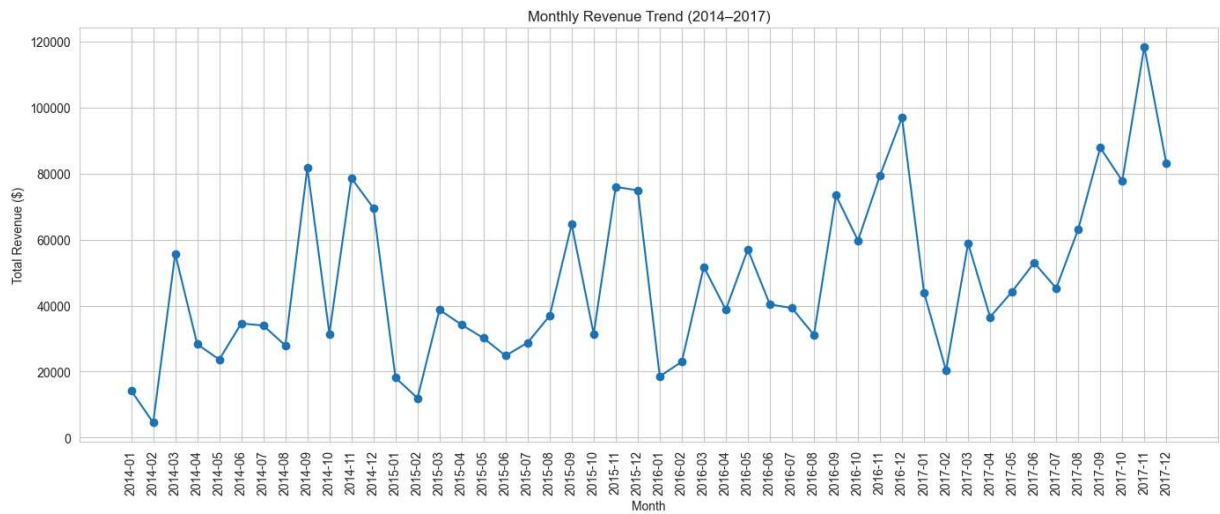


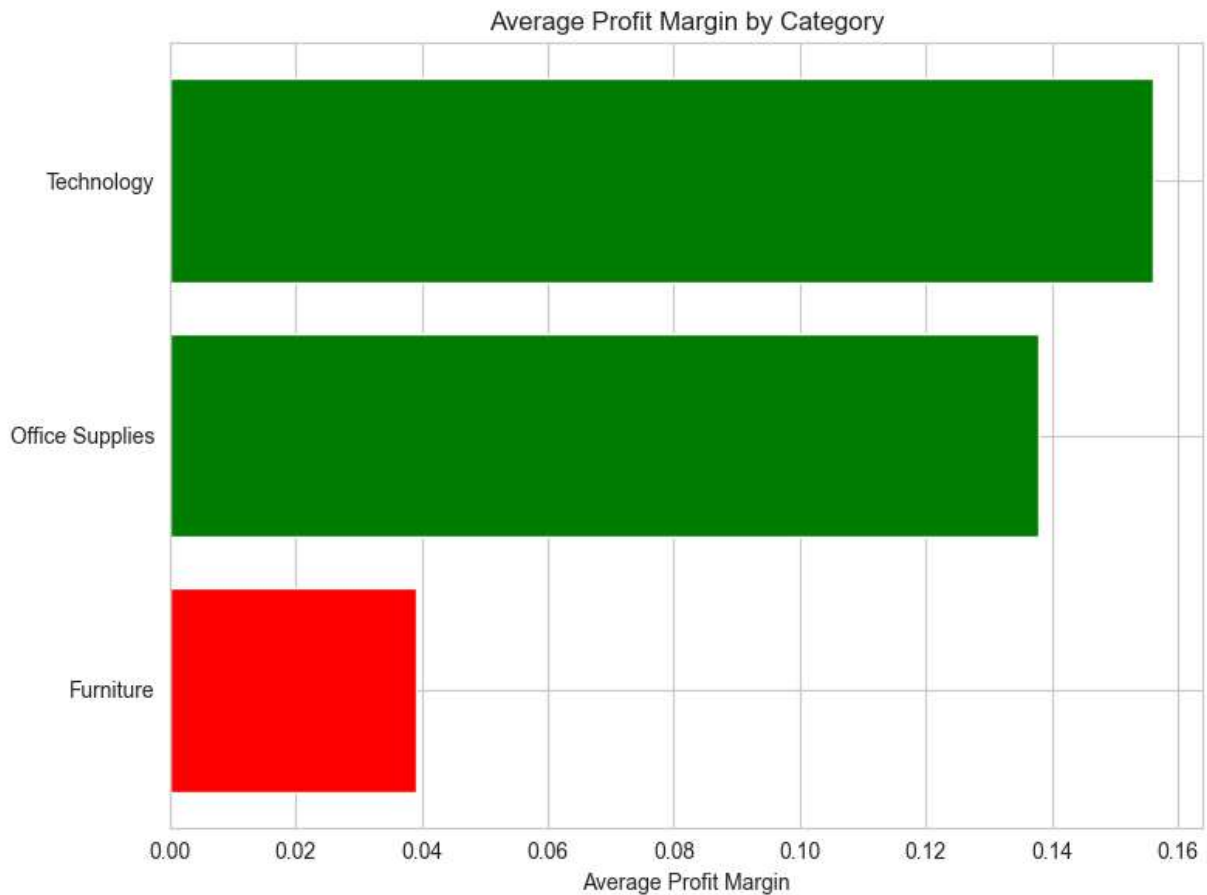
Chart 2: Profit Margin by Category

Highlights that Furniture has the weakest margin despite solid sales volume.

```
In [23]: category_margin = df.groupby('Category')['ProfitMargin'].mean().sort_values()

colors = []
for value in category_margin.values:
    if value < 0.1:
        colors.append('red')
    else:
        colors.append('green')

plt.figure(figsize=(8, 6))
plt.barh(category_margin.index, category_margin.values, color=colors)
plt.title('Average Profit Margin by Category')
plt.xlabel('Average Profit Margin')
plt.tight_layout()
plt.savefig('../dashboard/chart2_category_margin.png')
plt.show()
```



(Safety check) Recreate DiscountBand column

This column was created earlier in the first notebook. If it wasn't saved in the cleaned CSV, we recreate it here before using it in Chart 3.

```
In [24]: def discount_band(d):  
    if d == 0:  
        return 'No Discount'  
    elif d <= 0.2:  
        return 'Low (0-20%)'  
    elif d <= 0.4:  
        return 'Medium (20-40%)'  
    else:  
        return 'High (40%+)'  
  
    if 'DiscountBand' not in df.columns:  
        df['DiscountBand'] = df['Discount'].apply(discount_band)  
  
    print('DiscountBand' in df.columns)
```

True



Chart 3: Discount Band vs Average Profit

```
In [25]: band_order = ['No Discount', 'Low (0-20%)', 'Medium (20-40%)', 'High (40%+)']  
discount_profit = df.groupby('DiscountBand')['Profit'].mean().reindex(band_order)
```

```

colors = []
for value in discount_profit.values:
    if value > 0:
        colors.append('green')
    else:
        colors.append('red')

plt.figure(figsize=(9, 6))
plt.bar(discount_profit.index, discount_profit.values, color=colors)
plt.axhline(0, color='black')
plt.title('Average Profit per Order by Discount Band')
plt.xlabel('Discount Band')
plt.ylabel('Average Profit ($)')
plt.tight_layout()
plt.savefig('../dashboard/chart3_discount_vs_profit.png')
plt.show()

```

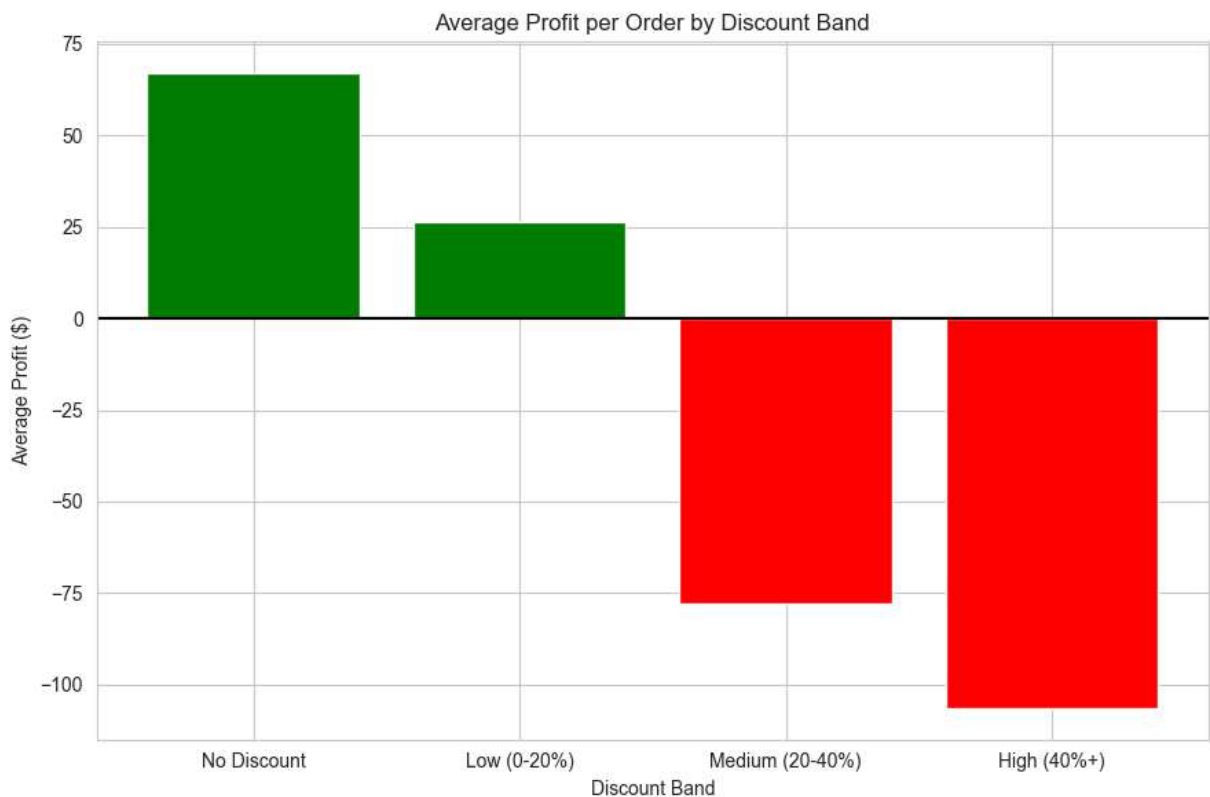


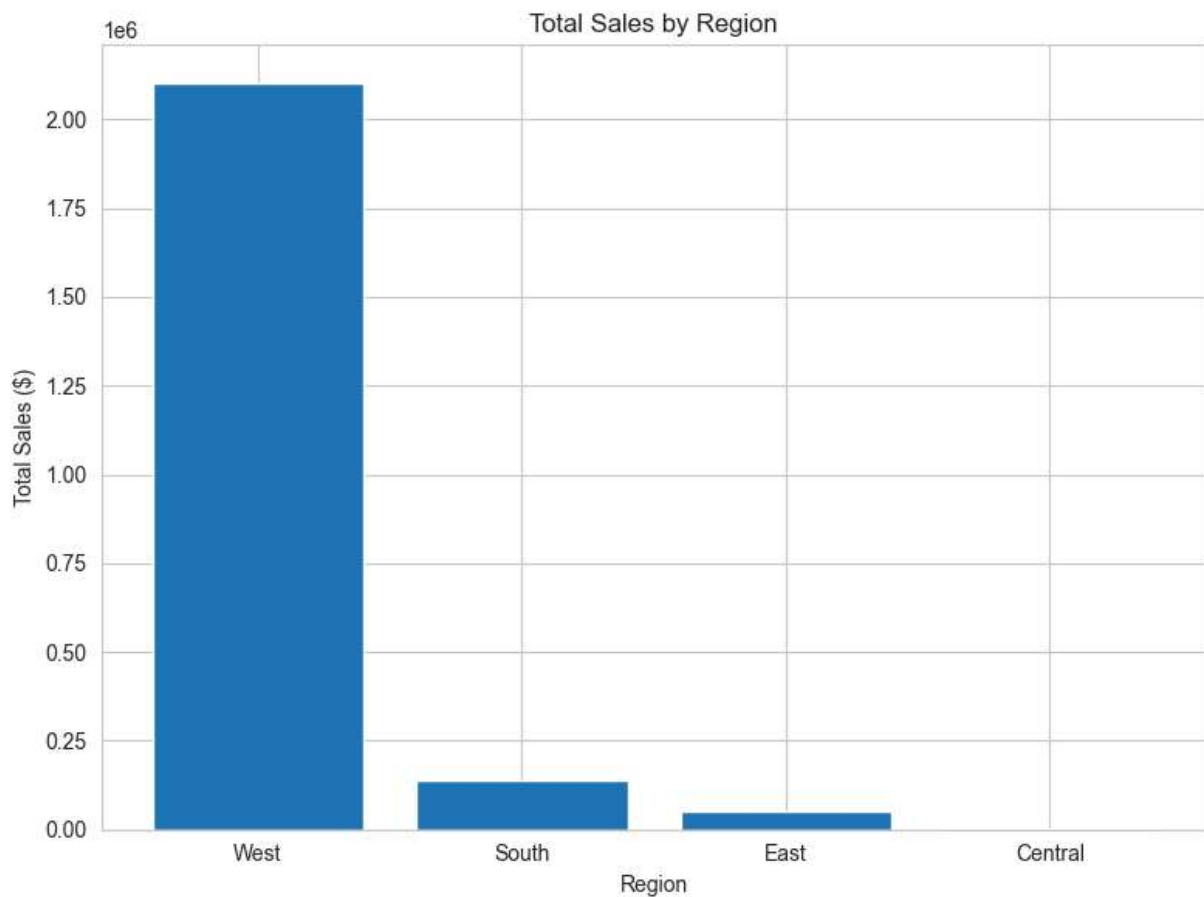
Chart 4 — Regional Sales Distribution

```

In [26]: region_sales = df.groupby('Region')['Sales'].sum().sort_values(ascending=False)

plt.figure(figsize=(8, 6))
plt.bar(region_sales.index, region_sales.values)
plt.title('Total Sales by Region')
plt.xlabel('Region')
plt.ylabel('Total Sales ($)')
plt.tight_layout()
plt.savefig('../dashboard/chart4_regional_sales.png')
plt.show()

```

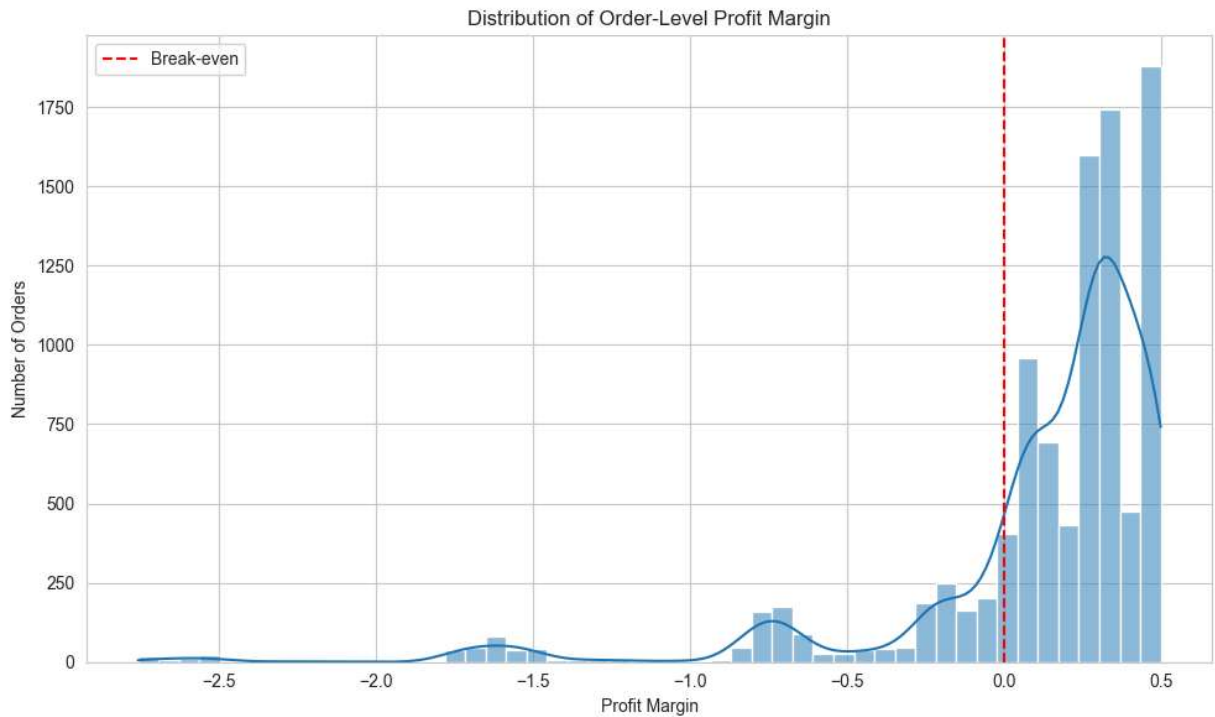


Finding: The West region accounts for the overwhelming majority of total sales (~2.1M), dwarfing South (138K), East (~52K), and Central (1.5K). This concentration suggests the business is heavily dependent on West region performance — a risk worth flagging for diversification.



Chart 5 — Profit Margin Distribution

```
In [27]: plt.figure(figsize=(10, 6))
sns.histplot(df['ProfitMargin'], bins=50, kde=True)
plt.axvline(0, color='red', linestyle='--', label='Break-even')
plt.title('Distribution of Order-Level Profit Margin')
plt.xlabel('Profit Margin')
plt.ylabel('Number of Orders')
plt.legend()
plt.tight_layout()
plt.savefig('../dashboard/chart5_margin_distribution.png')
plt.show()
```



Finding: The majority of orders fall between 0% and 50% profit margin, with a peak around 30-40% — indicating most of the business is healthy. However, there is a distinct left tail of loss-making orders, including a small cluster as low as -150% to -275% margin. These severe outliers likely correspond to the high-discount orders identified earlier and disproportionately pull down the overall average margin (12%) relative to the median (27%).

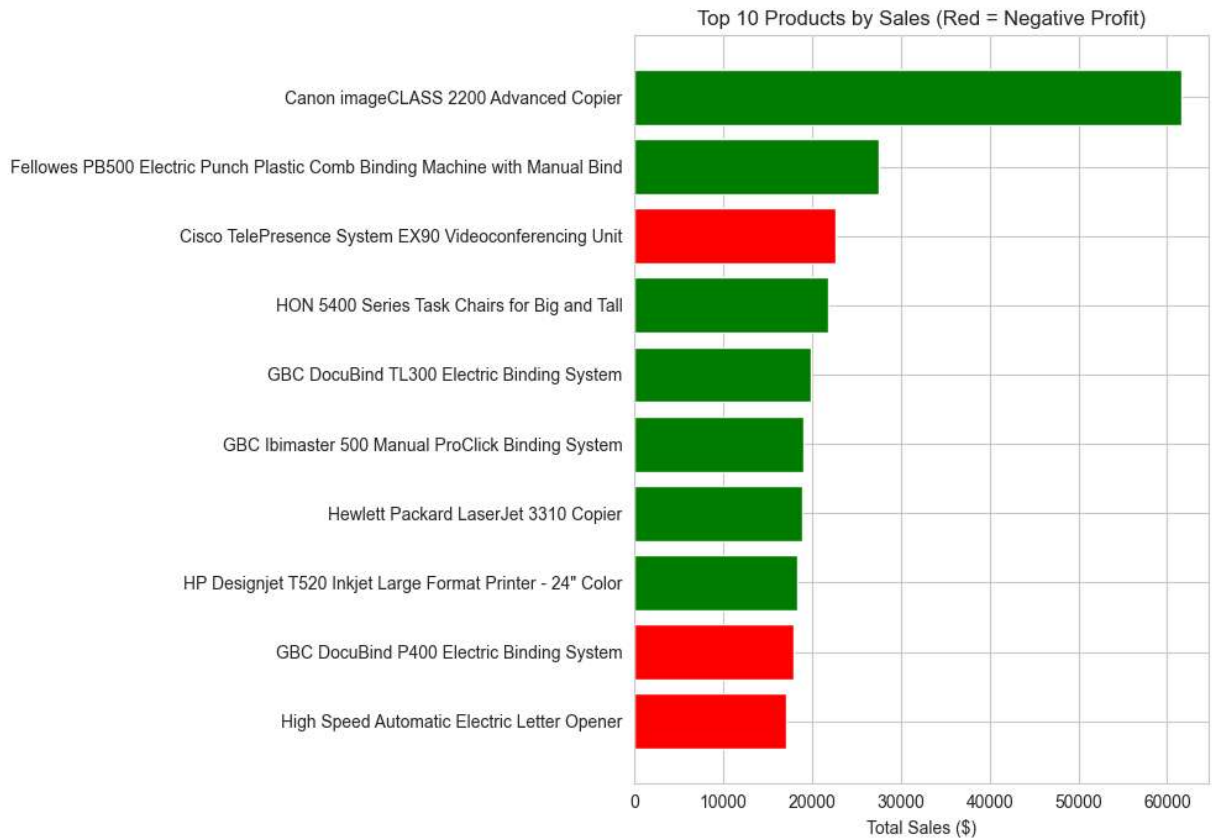


Chart 6 — Top 10 Products by Sales

```
In [28]: top_products = df.groupby('ProductName').agg(
    TotalSales=('Sales', 'sum'),
    TotalProfit=('Profit', 'sum')
).sort_values('TotalSales', ascending=False).head(10)

colors = []
for value in top_products['TotalProfit']:
    if value < 0:
        colors.append('red')
    else:
        colors.append('green')

plt.figure(figsize=(10, 7))
plt.barh(top_products.index[::-1], top_products['TotalSales'][::-1], color=colors[:])
plt.title('Top 10 Products by Sales (Red = Negative Profit)')
plt.xlabel('Total Sales ($)')
plt.tight_layout()
plt.savefig('../dashboard/chart6_top_products.png')
plt.show()
```



Finding: 3 of the top 10 best-selling products by revenue — Cisco TelePresence System, GBC DocuBind P400, and the High Speed Automatic Electric Letter Opener — are actually unprofitable. High sales volume does not guarantee profitability, reinforcing the need to evaluate products by margin, not just revenue.

Summary

This notebook validated the SQL analysis findings using Python, added two new insights (shipping delay has minimal profit impact; Furniture has a margin problem despite solid sales), and produced 6 visualizations covering seasonality, category margins, discount impact, regional distribution, margin spread, and product-level profitability.